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HOSPITAL SERIAN

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AMENDMENT PAGE

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INTRODUCTION

Infection prevention and control practices are important in maintaining a safe environment for patients by reducing the risk of the potential spread of disease from person to person.

Addressing infection prevention and control requires a facility wide programme and should be a priority in every healthcare institution, thus, infection control governance plays an integral role in patient healthcare.

1. ORGANIZATION AND MANAGEMENT

1.1

The Prevention and Control of Infection (PCI) Services shall be organized and administered to provide optimum support towards the implementation of safe infection control practices.

Vision, Mission and Values Statements

1.1.1 Vision Statement

Provide quality health services through preventive and control measures of infection and create a safe and clean environment to improve health status.

1.1.2 Mission Statement

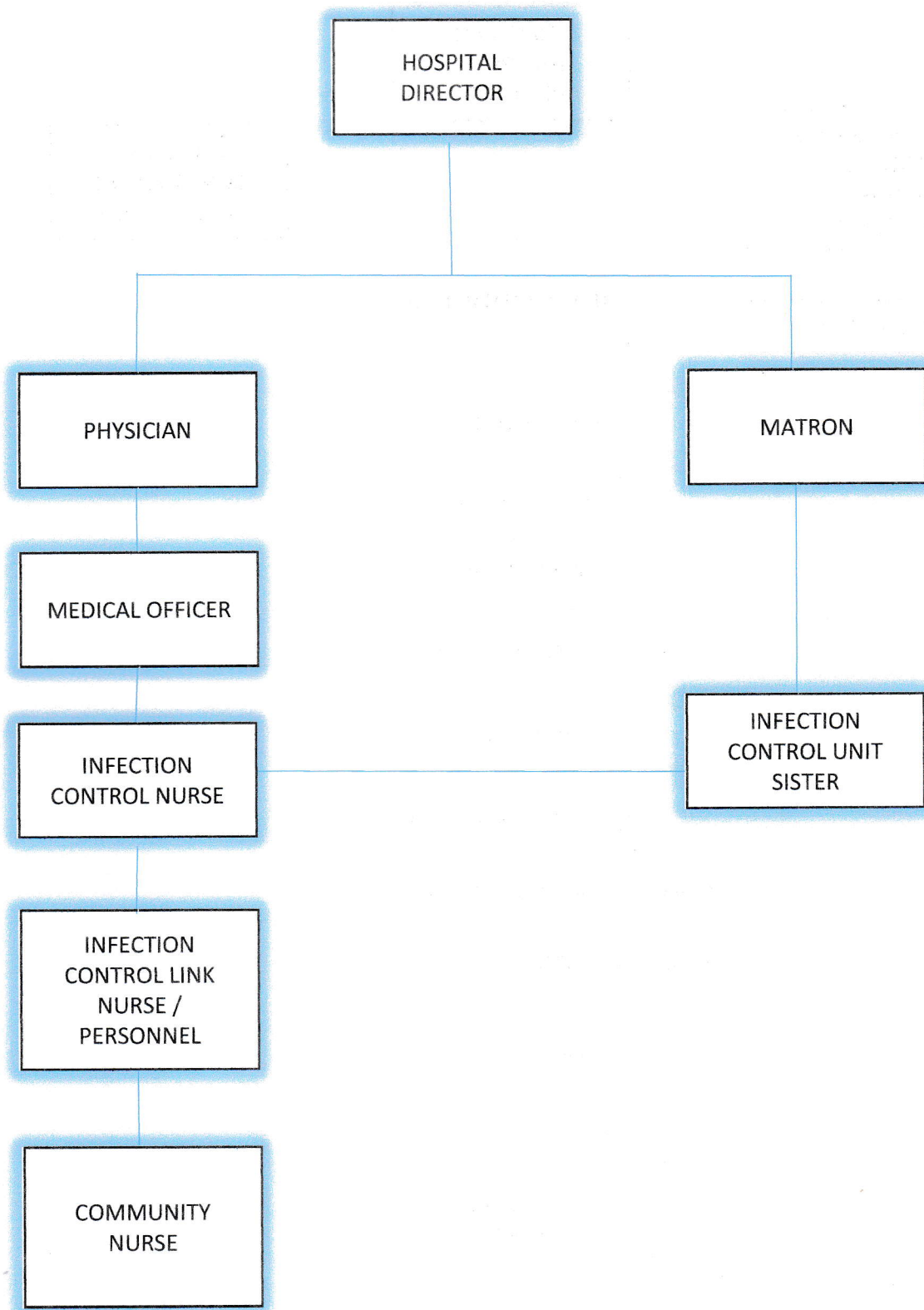
Ensure that individuals continue to adopt infection control practices and create an environment that is safe and free from healthcare associated infections.

1.1.3 Goals and Objectives

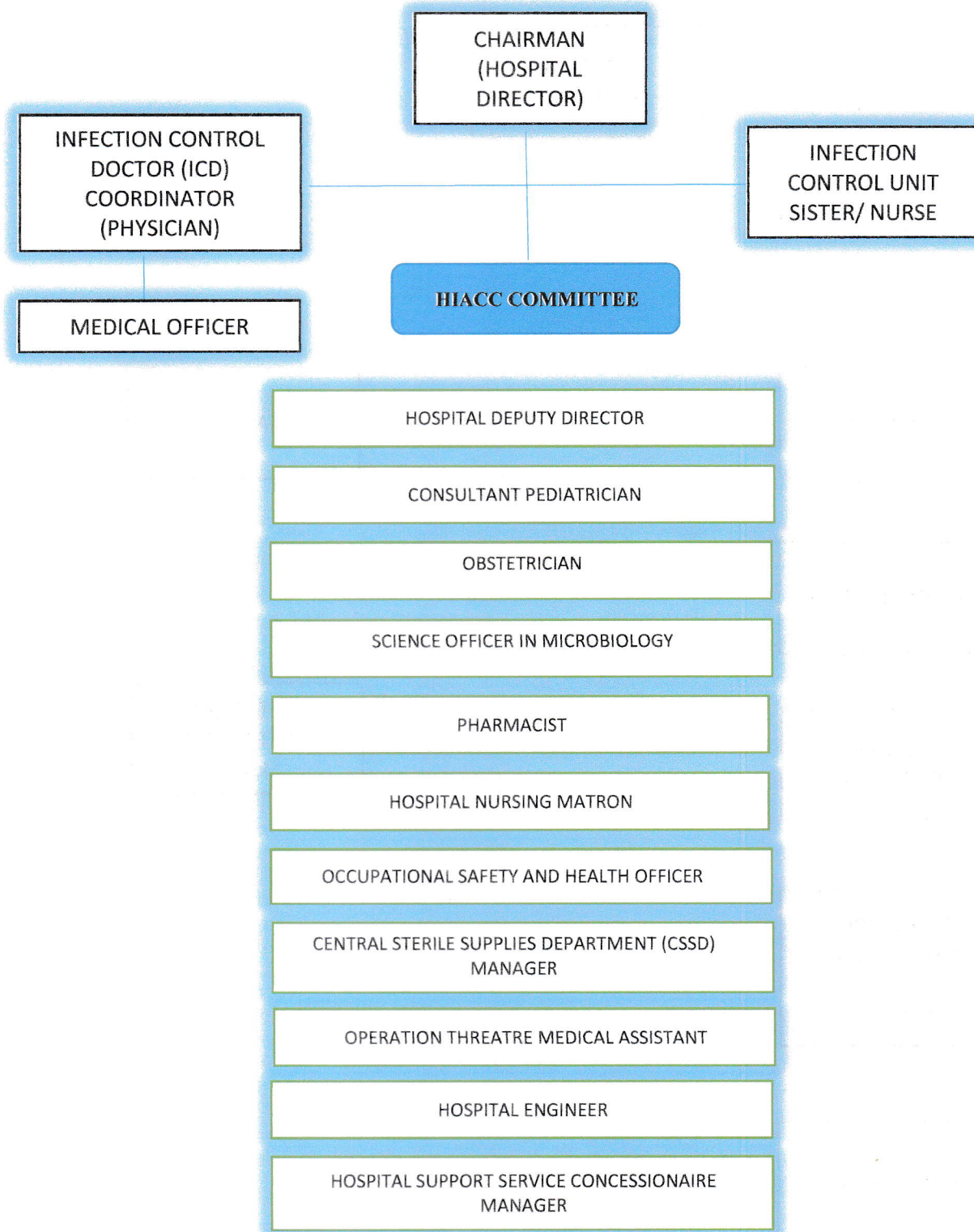
- I. To create a conducive and safe environment for the patient and healthcare workers.
- II. To promote awareness of the importance of hospital infection control for all categories of medical personnel through orientation programme and other education activities.
- III. To reduce the incidence of healthcare associated infection by promoting cost-effective hospital infection control practices.
- IV. To reduce the incidence of needle stick and sharp injuries in the hospital by promoting good occupational safety practices.

1.2 ORGANIZATION CHART

INFECTION CONTROL UNIT ORAGANIZATION CHART



HIACC ORAGANIZATION CHART



<u>1.3 HOSPITAL INFECTION AND ANTIBIOTIC CONTROL COMMITTEE (HIACC)</u>	
1.3.1	<u>GOVERNANCE</u> The HIACC functions as a governance body to coordinate hospital activities in prevention and control of HCAI, promote judicious use of antimicrobials and control the emergence of AMR. The HIACC involves a multidisciplinary participation from various departments and units.
1.3.2	<u>ROLES AND RESPONSIBILITIES</u> <u>1.3.2.1 Chairperson</u> i. Chairs the HIACC meetings. ii. Facilitates and coordinates the hospital infection prevention and antibiotics control activities. iii. Reviews and approves HIACC output. <u>1.3.2.2 Infection Control Doctor (ICD) Coordinator</u> i. Represent the hospital at SIACC meetings. ii. Assists chairperson in facilitation and coordination of the hospital infection prevention and antibiotic control activities. iii. Lead the hospital level infection prevention and antibiotic control activities. iv. Advise Chairman and Committee on actions to be taken if national performance indicators are not met (as set by NIACC). <u>1.3.2.3 Infection Control Medical Officer</u> i. Assists chairperson and ICD Coordinator in facilitation and coordination of the hospital infection prevention and antibiotic control activities. ii. Should have training and experience in matters relating to infection and antibiotic control. <u>1.3.2.4 Infection Control Sister / Nurse</u> i. Acts as a secretariat. ii. Schedules and prepares the agenda of each HIACC meeting. iii. Preparation of HIACC documents, reports and minutes of meeting. iv. Maintains a spreadsheet to track progress on recommendations. v. Facilitates HIACC correspondence. <u>1.3.2.5 Members of Hospital Infection Control and Antibiotic Committee</u> i. Representative from the department must be senior personnel ii. Participates in each HIACC meeting iii. Provides technical input

1.3.3	<u>TERM OF REFERENCE</u> <u>1.3.3.1 Information Sharing</u> The HIACC provides a structure for information sharing to mutually reinforce activities among department in the hospitals. <u>1.3.3.2 Interactions</u> The HIACC will interact with the health system and disease specific programme. Members of the HIACC are well represented and have clearly defined roles and responsibilities.
1.3.4	<u>FREQUENCY OF MEETINGS</u> 1.3.4.1 Meetings shall be held at least once in every four months. 1.3.4.2 Members of the committee shall be notified of the date and agenda of the meeting at least two weeks prior to the meeting. 1.3.4.3 Minutes should be kept and ratified.
1.3.5	<u>AGENDA OF THE MEETINGS</u> 1.3.5.1 Report on the incidence and prevalence of MDRO organisms and/or emerging resistant organisms. 1.3.5.2 Report on Hospital Surveillance of Antibiotic Resistance. 1.3.5.3 Report on Healthcare Associated Infection Surveillance. 1.3.5.4 Report on Antibiotic Utilization Surveillance. 1.3.5.5 Report on Infection Prevention and Control Audit. 1.3.5.6 Report on Hand Hygiene Compliance Surveillance. 1.3.5.7 Report on Sharp Injuries among Healthcare Worker. 1.3.5.8 Report on Tuberculosis among Healthcare Workers 1.3.5.9 Report on outbreak and action plan.
<u>1.4 PREVENTION AND CONTROL OF INFECTION (PCI) TEAM</u>	
1.4.1	<u>TERMS OF REFERENCE</u> i. Performing and reporting on surveillance for healthcare associated infection.

	ii. Investigating and advising on outbreak management. iii. Providing a clinical advisory service for infection prevention-related issues. iv. Revising and formulating policies. v. Providing regular in service training for all healthcare worker. vi. Reporting to and advocating for improved standards of care with facility management. vii. Auditing the quality and effectiveness of healthcare facility environmental cleaning, disinfection and sterilization process.
1.5	<u>INFECTION CONTROL LINK NURSE (ICLN) / LINK PERSONNEL</u> i. The ICLN is an appointed nurse in each ward with sufficient nursing experience preferable in infection control. ii. The ICLN acts as a liaison personnel between the HCWs in the ward and the Infection Control Team on matters related to infection control. <u>1.5.1 DUTIES AND RESPONSIBILITIES OF ICLN</u> 1.5.1.1 Supervision on Infection Control Practices i. Ensure hand hygiene is being practiced in the ward. ii. Ensure compliance with aseptic technique. iii. Ensure compliance with PPE. iv. Proper cleansing and sterilization according to standard procedures. v. Proper storage of sterile instruments and linen. vi. Proper collection and dispatch of specimens. vii. Proper segregation and disposal of healthcare waste. viii. Isolation of patients in accordance to transmission- based precautions. ix. Advise on immediate management of sharp objects. 1.5.1.2 Surveillance i. Participate in audit activities. ii. Assist in the prevention and reporting of sharp injuries among HCWs. iii. Assist in notification of outbreaks. 1.5.1.3 Education i. Acts as a resource personnel and advice on matters related to infection control. ii. Disseminate, educate and create awareness on infection control to HCWs in the ward.
1.6	1.6.1 HIACC meetings shall be held at least once in every four months and whenever necessary to discuss issues matters pertaining to the operations of the PCI services. 1.6.2 Minutes are kept; decisions and resolutions made during meetings shall be accessible, communicated to all staff of the service and implemented.

1.7	The chairperson of HIACC, who is the Head of Prevention and Control of Infection (PCI) Services, is involved in planning, justification and management of budget and resource utilization of the services.
1.8	The Head of PCI is involved in the assignment of staff.
1.9	Appropriate statistics and records shall be maintained in relation to the provision of PCI Services and used for managing the services and patient care purposes.
1.10	1.10.1 Where more than one committee have interests in the issues of the PCI Services, there is evidence of coordination of the actions undertaken or proposed by the committees. 1.10.2 Records are kept on actions taken to identify and correct the cause of any problem.
1.11	1.11.1 There are safety measures taken to ensure the protection of the Facility's staff and environment against healthcare associated infections. 1.11.2 Records shall be kept on action taken which include: i. Staff education; ii. Staff health screening including infectious diseases; iii. Staff immunization; iv. Staff health record maintenance; v. Provision for adequate and good quality personal protective equipment (PPE); vi. Implementation of safety devices; vii. Clinical waste management; (Refer Appendix 1 Clinical Waste management) viii. Protocol for post exposure management for infectious disease and for assignment of infected staff. (Refer Sharp Injury Surveillance Manual MOH 2007)
1.12	Provision is made for the personal comfort and safety of patients and staff which include: i. Clean and hygienic facilities; ii. Room temperatures are kept at comfortable levels; iii. Disinfection and sterilization areas; safe equipment and instruments; iv. Proper hand hygiene facilities; v. Aseptic techniques for procedures; vi. Practice of standard and additional precautions.

2. HUMAN RESOURCE DEVELOPMENT AND MANAGEMENT

2.1 The head and staff of the PCI Services shall be individuals qualified by education, training, experience and certification to commensurate with the requirements of various positions.

2.2 **ROLES AND RESPONSIBILITIES OF HEAD OF PCI SERVICES**

- i. Organizing surveillance for healthcare associated infections.
- ii. Providing advice and leadership in outbreak investigation.
- iii. Developing and delivering training to healthcare workers.
- iv. Developing and implementing PCI related policies and procedures.
- v. Auditing the quality and effectiveness of healthcare facility environmental cleaning.
- vi. Auditing the quality and effectiveness of disinfection and sterilization process.

2.3 The infection control nurse (ICN) in-charge shall have post basic infection control training and certified.

2.4 The Infection Control Nurse in-charge has delegated authority for the supervision and effective implementation of infection control policies, and is responsible for surveillance of healthcare associated infections on a systematic and current basis.

2.5 Sufficient numbers of personnel and support staff including link nurses / link personnel with appropriate qualifications are employed to meet the need of the services (National norms is 1 ICN :110 beds).

2.6 **ROLES AND RESPONSIBILITIES FOR PCI TEAM****2.6.1 CLINICAL DUTIES**

- i. Work closely with other members of HIACC and PCI team.
- ii. Supervise and advice on infection prevention and control policies.
- iii. Provide clinical advice and support to HCW.
- iv. Determine the clinical significance of laboratory results of MDROs or possibility of an outbreak.
- v. Provide guidance and support to the HCWs.

2.6.2 SURVEILLANCE

- i. Coordinate surveillance activities for the hospital
- ii. Collect relevant information including point prevalence studies on HCAI, antibiotic audits, hand hygiene compliance etc

2.6.3 COORDINATION / ORGANIZATION OF INFECTION CONTROL ACTIVITIES

- i. Identify potential infectious hazards and suggest appropriate remedial action to relevant personnel.

	ii. Recognize, investigate and implement control measures during outbreaks. <u>2.6.4 ADMINISTRATIVE</u> i. Participate in the development and implementation of the infection control policies. ii. Monitor compliance with infection control policies, including activities directly related to the audits. <u>2.6.5 EDUCATION</u> i. Participate in teaching programmes for all HCWs. ii. Keep abreast with recent advances by reading relevant literature and attending appropriate courses, meetings and exhibitions. iii. Participate and coordinate infection control related educational campaigns <u>2.6.6 QUALITY IMPROVEMENT ACTIVITIES</u> i. Perform clinical audit / quality improvement projects on infection control activities and to evaluate its effectiveness.
2.7	2.7.1 All new staff are required to attend structured orientation programme 2.7.2 Orientation module 2.7.2.1 Fundamental Principles of Infection Prevention i. <u>Standard precautions</u> a. Hand hygiene b. Personal Protective Equipment (PPE) c. Disinfection and Sterilization d. Environmental hygiene e. Waste management f. Linen management g. Spillage management h. Injection safety and Sharps management i. Respiratory hygiene and cough etiquette ii. <u>Transmission Based Precautions</u> a. Contact b. Droplet c. Airborne d. Translocation
2.8	Healthcare facilities should provide specific education and training for all healthcare workers about infection prevention and control policies and procedures which provide the knowledge and skills required for staff to maintain competency in their current positions and future advancement.

Infection Control Unit Operational Policy

2.9	There are continuing education activities for staff to pursue professional interests and to prepare for current and future changes in practice.
2.10	All staffs have the opportunity to attend additional training in the conduct of procedures unique to the services such as the operating rooms, obstetrical units, emergency services, special care units, central sterilising supply services and isolation rooms etc.
2.11	The PCI team is involved in orientation and in service education for all staff .

3. POLICIES AND PROCEDURES**3.1**

3.1.1 Facility-wide infection control policies and procedures which are customized to the complexity of the services provided. The policies and procedures address the following:

3.1.2.1 Preventive and control procedures for all aseptic techniques and practices related to sterilization and disinfection; (**Refer Appendix 2**)

3.1.2.2 Fight against antimicrobial resistance; (**Refer Antimicrobial Policy Hospital Serian**)

3.1.2.3 Use of personal protective equipment (PPE) (**Refer Appendix 3**)

3.1.2.4 Cross-infection and isolation (patients and visitors) (**Refer Appendix 4**)

3.1.2.5 Central sterilising supply services; (**Refer Policy CSSD Hospital Serian**)

3.1.2.6 Housekeeping services; (**Refer Appendix 5**)

3.1.2.7 Laundry services; (**Refer Appendix 6**)

3.1.2.8 Food handling; (**Refer Polisi, Manual Dan Prosedur Unit Sajian Hospital Serian**)

3.1.2.9 Handling of sharps and waste; (**Refer Appendix 1**)

3.1.2.10 Pharmacy services; (**Refer Hospital Serian Pharmacy Service Policy 2020**)

3.1.2.11 Pathology Services; (**Refer Pathology Lab Policy Hospital Serian 2020**)

Refer to Policies and Procedures on IPC, MOH Third Edition)

3.1.2.12 Surgical and nursing procedures;

3.1.2.13 Engineering services;

3.1.2.14 Ventilation system;

3.1.2.15 Facility and equipment maintenance, and all others

3.2	Policies and procedures are developed by a committee in collaboration with staff, medical practitioners.
3.3	Current guidelines, policies and procedures are communicated to all staffs.
3.4	Healthcare worker compliance with policies and procedures and evidence based guidelines (World Health Organization / Centre for Disease Control and Prevention / Ministry of Health) as stated in 3.1.
3.5	Copies of policies and procedures, protocols, guidelines, relevant acts, regulations, by-laws and statutory requirements are accessible to staff.
3.6	Current reference manuals, pamphlets, journals, and books as well as information and scientific data concerning infection control shall be available for reference and guidance.
3.7	Regular "environmental infection risks" inspections are conducted throughout the Facility for the purpose of updating policies and procedures related to infection control practices.
3.8	The HIACC shall be consulted in order to ensure that: i. Proposed demolition, building constructions and renovations are designed in line with accepted infection control requirements; ii. Proposed new equipment intended for patient care conforms to accepted infection control standards.
3.9	3.9.1 The HIACC reviews reports on healthcare associated infection rates, surveillance studies of infections and infection potentials, and the implementation of infection control policies. 3.9.2 Pertinent findings shall be submitted to the appropriate source for necessary action.
3.10	Policies and procedures for infectious patients and those requiring isolation and treatment are available and complied including the following: i. Proper isolation of infectious cases for patients with compromised immune systems and airborne infection and newborns; (Refer Appendix 4) ii. Hand hygiene practices; iii. The isolated patients receive the same quality of care as is provided throughout the Facility.

4. FACILITIES AND EQUIPMENT	
4.1	There are adequate and appropriate facilities and equipment with proper utilization of space for patient management to enable staff to carry out their professional and administrative functions.
4.2	4.2.1 The use of all medical devices and disinfectants shall comply with the manufacturers' instructions on prevention and control of infection. 4.2.2 Refer to <i>Occupational Safety And Health (Classification, Labelling And Safety Data Sheet of Hazardous Chemicals) Regulations 2013</i>
4.3	There shall be no reprocessing of any single-use medical-surgical instruments, equipment if there are adequate supplies.
4.4	Adequate personal protective equipment (PPE) shall be provided for healthcare providers, patients and visitors where appropriate.
4.5	Isolation facilities for airborne infection and immunocompromised patients shall comply with regulatory requirements.
4.6	Adequate and appropriate hand hygiene facilities including alcohol based hand rub shall be available in all patients, staff and visitor areas.

5. SAFETY AND PERFORMANCE IMPROVEMENT ACTIVITIES	
5.1	5.1.1 There are planned and systematic safety and performance improvement activities to monitor and evaluate the performance of the PCI Services. 5.1.2 The process includes: a) Planned activities; b) Data collection; c) Monitoring and evaluation of the performance; d) Action plan for improvement; e) Implementation of action plan; f) Reevaluation for improvement;
5.2	The Head of PCI Services has assigned the responsibilities for planning, monitoring and managing safety and performance improvement activities to appropriate individual / personnel / committee within the respective services.
5.3	5.3.1 The Head of PCI Services shall ensure that the staff are trained and complete incident reports which are promptly reported, investigated, discussed by the staff with learning objectives and forwarded to the Person In Charge (PIC) of the Facility. 5.3.2 Incidents reported have had Root Cause Analysis done and action taken within the agreed time frame to prevent recurrence.
5.4	5.4.1 There is tracking and trending of specific performance indicators of the following: <u>5.4.1.1 Percentage of staff trained in Prevention and Control of Infection Practices</u> Target: i. 100% new staff given orientation on infection control; ii. 85% of existing staff trained on infection control <u>5.4.1.2 Percentage of healthcare associated infection</u> Target: i. Healthcare associated infection rate: < 5% ii. Clinical based surveillance: - Surgical Site Infection (e.g. Clean operation less than 2%) - Urinary Tract Infection (UTI) < 5% - Catheter related infection (e.g. CRBSI) - < 5 per 1000 catheter days iii. Laboratory related surveillance: - Methicillin-resistant <i>Staphylococcus aureus</i> (MRSA) - (Target 0.3%) - Extended spectrum beta-lactamase (ESBL) producers - (Target 0.3%) - Carbapenem-resistant <i>Enterobacteriaceae</i> (CRE) - (Target 0.3%) - Vancomycin-resistant <i>enterococci</i> (VRE) - (Target 0.3%)

5.5	The Facility takes appropriate actions on all reportable diseases, emerging and re-emerging diseases and reports to the relevant authorities accordance to regulatory requirement. NOTE: Prevention And Control Of Infectious Diseases Act 1988.
5.6	Feedback on results of safety and performance improvement activities are regularly communicated to the staff and relevant authority.
5.7	Appropriate documentation of safety and performance improvement activities is kept and confidentiality of medical practitioners, staff and patients is preserved.

APPENDICES

APPENDIX 1

WASTE MANAGEMENT

Waste generated by health care activities includes a broad range of materials, from needles and syringes to soiled dressings, body parts, diagnostic samples, blood, chemicals, pharmaceuticals, medical devices and radioactive materials.

Of the total amount of waste generated by healthcare activities, about 85% is general, non hazardous waste comparable to domestic waste. The remaining 15% is considered hazardous material that maybe infectious, chemical or radioactive.

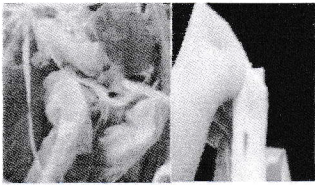
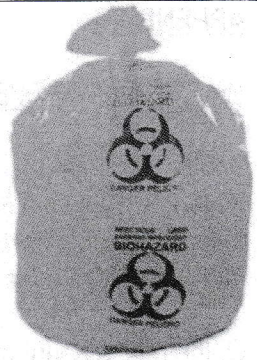
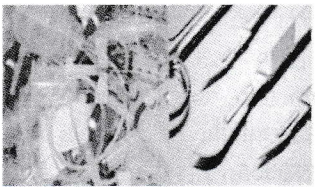
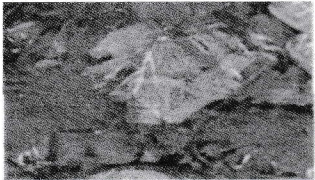
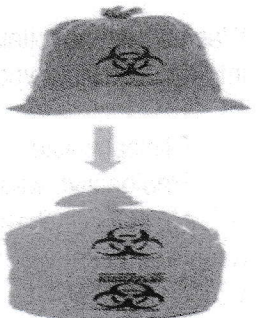
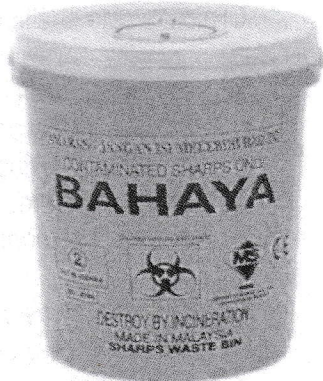
Poor management of health care waste potentially exposes health care workers, waste handlers, patients and the community at large to infection, toxic effects and injuries, and risks polluting the environment. It is essential that all medical waste materials are segregated at the point of generation, appropriately treated and disposed of safely.

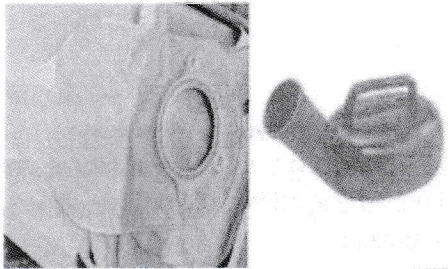
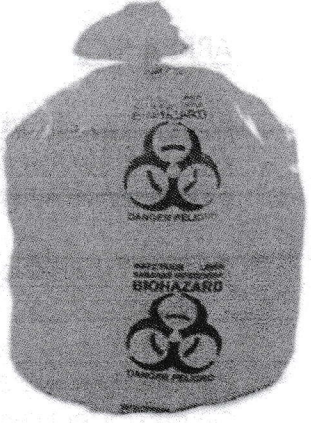
Wastes from hospitals and healthcare establishments can be categorized into the following types:

- i. Clinical waste
- ii. Radioactive waste
- iii. Hazardous chemical waste
- iv. Pharmaceutical waste
- v. General waste

Clinical waste shall be defined as:

- Any waste which consists of wholly or partly of human or animal tissues, blood or other body fluids, excretions, used rug, swabs dressings, syringes, needles or other instruments, being waste which unless rendered safe may prove hazardous to any other person coming into contact with it; and
- Any other waste arising from medical, laboratory, nursing, dental, investigation, treatment care, teaching or research, or the collection of blood for transfusion, being waste which cause infection to any other person coming into contact with it.

TYPES	EXAMPLE	WASTE SEGREGATION
CLINICAL WASTE GROUP A 	- waste contaminated with blood and other bodily fluids(e.g. Soiled gloves, plasters, swab, cotton wool, dressing, bandages etc), - pathological waste including all human tissue, organ, body parts, placenta, tissue from laboratories etc.	
CLINICAL WASTE GROUP B 	- Sharp waste E.g. Syringes, needles, disposable scalpels and blades and other sharp instruments that could cause a cut or puncture Trocar is recommended to be discarded into a well fit trocar container.	
CLINICAL WASTE GROUP C 	Waste from laboratory E.g. Used culture plates, tubes, bottles, blood culture bottles	
CLINICAL WASTE GROUP D 	Pharmaceutical waste E.g. Expired, unused and contaminated drugs and vaccine.	

CLINICAL WASTE GROUP E	Disposable medical instruments (non sharp) E.g. Bed pan, stoma bag, urine container 	
GENERAL WASTE (non clinical/ non hazardous waste)	Waste that does not pose any particular infectious/ biological, chemical, radioactive or physical hazard.	

APPENDIX 2

POLICY ON PREVENTIVE AND CONTROL PROCEDURES FOR ALL ASEPTIC TECHNIQUES AND PRACTICES RELATED TO STERILIZATION AND DISINFECTION

Aseptic Technique

The aseptic technique is a method to prevent transmission of microorganisms from various sources to a patient by creating a microorganism-free environment, maintaining sterility of instruments and preventing microbial contamination during various clinical procedures performed on a patient.

2.1 Non touch technique

- Non touch technique is a method of performing a procedure without directly touching any other surface that might come into contact.

2.2 Minimizing microorganisms on hands by hand hygiene

- Hand hygiene is a must before and after performing any clinical procedure. This practice maintains the cleanliness of the health care workers' hands at all times.

2.3 Rendering the Hand Sterile by Wearing Sterile Gloves

- Sterilized gloves are worn to render the hand sterile.
- If the glove is punctured or torn, replace it.

2.4 Minimizing bacteria at entry points

- The presence of bacteria depends on the site where the procedure is to be performed.
- The patient's skin harbour commensals (e.g. *Staphylococcus epidermidis*), which are harmless on the skin surface but may cause infection if it enters the bloodstream.
- Use antiseptic for skin preparation prior to sterile procedures.

2.5 Creating a sterile field

- There should be a sterile area within which instruments used for the intended procedure can be placed without risk of it being contaminated.

2.6 Maintaining sterility of instruments / disposable items during a procedure

APPENDIX 3

PERSONAL PROTECTIVE EQUIPMENT

Personal protective equipment, commonly referred to as “PPE”, is equipment worn to minimise exposure to hazards that cause serious workplace injuries and illnesses.

These injuries and illnesses may result from contact with chemical, radiological, physical, electrical, mechanical, or other workplace hazards. Personal protective equipment may include items such as gloves, safety glasses and shoes, earplugs or muffs, hard hats, respirators, or coveralls, vests and full body suits.

Recommendation by Occupational Safety and Health Administration (OSHA) United States

When selecting PPE, three key things need to be considered:

- **Type of anticipated exposure.**

This is determined by the type of anticipated exposure, such as touch, splashes or sprays, or large volumes of blood or body fluids that might penetrate the clothing and by the category of isolation precautions a patient is on.

- **Durability and appropriateness of the PPE for the task.**

Whether a gown or apron is more suitable. If a gown is selected, whether it needs to be fluid resistant, fluid proof, or neither.

- **Fit.**

PPE must fit the individual user, and the employer should ensure that all PPE are available in sizes appropriate for the workforce that must be protected.

All PPE should be:

- Located close to the point of use;
- Stored to prevent contamination in a clean/dry area until required for use (expiry dates must be adhered to);
- Single use items unless specified by the manufacturer; and
- Disposed of after use into the correct waste stream i.e.. healthcare waste or domestic waste.
- Reusable PPE items, e.g. non-disposable goggles/face shields/visors must have a decontamination schedule with responsibility assigned.

Gloves

Glove wearing by HCWs is recommended for two main reasons:

1. To prevent microorganisms which may be infecting, commensally carried, or transiently present on HCWs' hands from being transmitted to patients and from one patient to another; and

3. To reduce the risk of HCWs acquiring infections from patients.

Gloves must be:

- Worn when exposure to blood and/or other body fluids is anticipated/likely.
- Changed immediately after each patient and/or following completion of a procedure or task;
- Changed if a perforation or puncture is suspected; and
- Appropriate for use, fit for purpose and well-fitting to avoid excessive sweating and interference with task performance.

Isolation Gowns and Aprons

1. Disposable plastic aprons should be worn when there is a risk that clothing or uniform may become exposed to blood, body fluids, secretions and excretions.
2. Full body gowns need only be used where there is the possibility of extensive splashing of blood, body fluids, secretions or excretions and should be fluid repellent.
3. However, when contact precautions are used to prevent transmission of an MDRO, donning of both gown and gloves prior to room entry, regardless of the anticipated level of contact, may reduce unanticipated contact with an MDRO in the environment.
4. Removal of isolation gowns before leaving the patient care area is advised to prevent opportunities for possible contamination outside the patient's room.

Face Protection: Masks, Goggles, Face Shields

1. Masks are used for three primary purposes in healthcare settings:
 - a. to protect health care workers from contact with infectious material from patients e.g. respiratory secretions and sprays of blood or body fluids as defined in standard and droplet precautions.
 - b. worn by healthcare workers when engaged in procedures requiring sterile technique to protect patients from exposure to infectious agents carried in a healthcare worker's mouth or nose.
 - c. placed on coughing patients to limit potential dissemination of infectious respiratory secretions from the patient to others (i.e.. Respiratory Hygiene/ Cough Etiquette).
2. Two types of mask available, the surgical mask and particulate respirator (e.g. N95) used to prevent inhalation of small particles that may contain infectious agents transmitted via the airborne route.
3. Disposable or non-disposable face shields may be used as an alternative to goggles. As compared with goggles, a face shield can provide protection to other facial areas in addition to the eyes.

CAPS, BOOTS AND SHOE COVERS

Wear caps and boots / shoe cover where there is a likelihood the patient's blood, body fluid, secretions, excretions may splash, spill or leak onto the hair or shoes.

APPENDIX 4

ISOLATION OF INFECTIOUS CASES

- ✓ The laboratory shall inform the Infection Control Unit and the ward when a patient is identified.
- ✓ If a patient is admitted with a resistant organism, the nursing staff shall inform the Infection Control Unit.
- ✓ Patients with MRSA should ideally be isolated in a single room and Standard Precaution measures carried out.
- ✓ If no room is available, then patient should be placed at the end corner of the ward, with a sign at the door or screen to alert staff.
- ✓ Gowns and gloves are required for all patients coming in close contact with the patient. Other PPE should be readily available and should be used as necessary.
- ✓ Hand hygiene and use of PPE must be strictly enforced. Chlorhexidine soap is use for routine hand hygiene and hand rub may be use if hands are not visibly dirty.
- ✓ Specific HCW may be dedicated to provide care for the patients. Number of staff entering the infected patient area should be kept to a minimum.
- ✓ Visitors to be restricted and advised on PPE and hand hygiene.
- ✓ Patients and family must be inform regarding the need for isolation and control measures.
- ✓ Transport and movement of the patient are kept to a minimum, when necessary only and carefully supervised.
- ✓ Patient for surgery should be scheduled for the end of the operation session.
- ✓ Appropriate barrier (e.g. mask, impervious dressings) are to be taken to reduce the transmission of organisms to other patients, HCW or the hospital environment. Dressings and linen must be changed prior movement to avoid cross contamination.
- ✓ Personnel at the receiving end must be notified in advance and of the precautions to be taken.
- ✓ Patient care equipment and other articles required are to be carried into room only when required.
- ✓ Where possible, there should be dedicated use of non critical items. If use of common items is unavoidable, then it must be properly cleaned and disinfect before being use on other patients.

- ✓ Wheelchairs and stretcher used must be decontaminated after each use.
- ✓ All linen are to be treated as contaminated and should be discarded into red bag kept in the room or patient area. HCW must wear appropriate PPE to prevent soiling of their clothes.
- ✓ Patient's immediate environment, bed-rails, over bed table, bedside tables etc must daily and more frequently if needed. This should be done last, using a dedicated set of cleaning equipment which should decontaminated after use.
- ✓ All waste/ trash generated are considered as contaminated and discarded into clinical bin in the room.
- ✓ All reusable items must be decontaminated after patient discharged or taken out of isolation.
- ✓ Mortuary staff must be inform of infectious status before corpse is transfer to the mortuary.

APPENDIX 5

HOUSEKEEPING SERVICES

- ✓ Ensure procedures for the routine care, cleaning, and disinfection of environmental surfaces, beds, bed-rails, bedside equipment and other frequently touched surfaces, are being followed and kept clean and dust free
- ✓ All patient area should be clean and dry. All surfaces, fixture, furniture are free from dust or dirt.
- ✓ Bedpans and urinals must be washed clean and disinfected in bedpan sterilizer for at least 1 minute at 80 degree Celsius.
- ✓ Bathroom / toilet / sluice and all sinks must be washed at least daily or PRN when dirty.
- ✓ Treatment room must be clean and dry.
- ✓ Decorations of wards such as flowers are not recommended as it is a source of collection of dust.

Colour coding of mop heads and cleaning cloth used in the cleansing services: RED, BLUE, GREEN, YELLOW and WHITE

APPENDIX 6

LINEN MANAGEMENT

GENERAL PRICIPLES

a) Hygienic clean linen supply

- ✓ The employees handling the hygienic clean linen must be well trained.
- ✓ The employees or HCW should follow proper hand hygiene procedures and sanitize prior to handling clean healthcare linens and after touching potentially contaminated surfaces.
- ✓ They should don appropriate PPE (apron and mask) but gloving is not required.

i) Storage

Requirements for the storage area:

- Clean, not dusty, free of vermin and away from drains and pipes.
- Temp to 20-25 degree Celsius (in the air-conditioned area).
- Linen cupboard must always be kept closed to reduce dust accumulation.
- Cupboard and shelves must not be made of wooden material. The materials must be solid and easily cleaned.
- Shelves in the storage should be 5 cm from the walls, 20-25 cm from the floor, 50 cm from the ceiling. "First in, first out" (FIFO) must be adhered to based on date of supply (date must be stated on the packet).
- Regular cleaning schedules (daily) of the cupboard and shelves must be done and documented regularly.
- Clean linen must never be placed on the floor or contaminated surfaces.
- Linen storage facilities are to be used for the storage of clean linen items only and not used for storage of any other equipment.
- The linen that does not meet the clean hygienic standards should be rejected and returned.

ii) Transportation and Distribution

- The hygienically clean linen should be wrapped before transportation from the laundry plant but need to avoid plastic wrapper in view of local humidity and temperature.
- Carts transporting hygienic clean linen from laundry to medical facilities must be clean.
- The clean linen must be covered during the transport to the central storage and delivery to the wards/units.
- Carts used for clean or soiled linen must be separated with physical barriers.
- If clean linen is taken into an isolation area and not used the linen must be laundered again before storage/use.
- Staff member should avoid pressing/hugging hygienic clean linen to their uniforms at any times.

Soiled / dirty linen

Contamination of clothing and transfer of micro-organisms to other patients and the environment should be avoided. Adherence to standard precautions when handling contaminated linen and minimizing agitation of the contaminated items are considered sufficient to prevent the dispersal of potentially infectious aerosols.

(Appropriate colour coded linen bags)

White linen bag :- For used (ordinary soiled) linen.

Red linen bag with alginate bag :- Linen contaminated with blood and body fluids and infected linen.

Green linen bag :- For theatre linen.

Brown bag :- For reject linen.